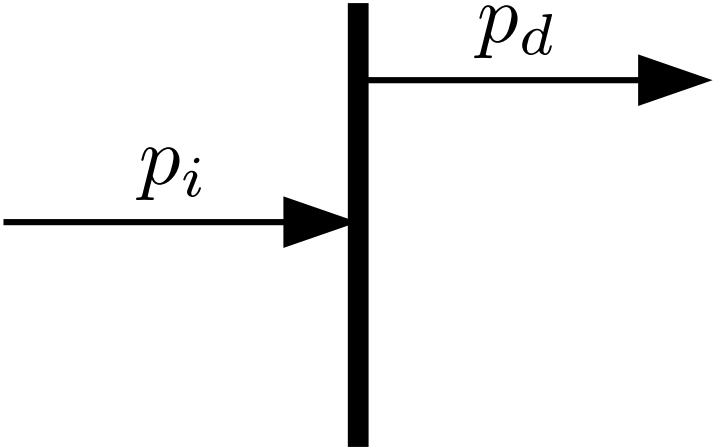
$p_d$
$p_i$